Angular Modules

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Key Learning Points

- What are Angular Modules
- How to implement an Angular module
- Properties of @NgModule
- Concept of Root Module
- Feature Modules

Modules in Angular Application

UserModule

- ProfileComponent
- RegComponent

ProductsModule

- Products Component
- Cart Component

RootModule

- LoginComponent
- MenuComponent

Angular Modules

- Angular apps are modular and Angular has its own modularity system called Angular Modules or NgModules.
- Modules help organize an application into cohesive blocks of functionality.
- Every Module can consist of multiple components.
- A Module acts as a container for multiple components, directives, pipes and services.
- Module files are named with **.module** suffix.

How to implement a Module?

- Module is a simple TypeScript class.
- The class has the @NgModule annotation.
- Component files are named with .module suffix.

@NgModule

NgModule is a decorator function that takes a single metadata object whose properties describe the module.

1. **Imports**	List of other modules required by the application.
2. **Declarations**	List of custom components, pipes and directives.
3. **Bootstrap**	Name of Root component(s).
4. **Providers**	List of services injected in the Root component.

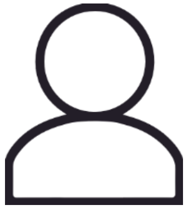
Declarations

- You tell Angular which components belong to the Module by listing it in the module's declarations array.

```
@NgModule({
  imports:      [ BrowserModule ],
  declarations: [ AppComponent,WelcomeComponent ],
  bootstrap:    [ AppComponent ]
```

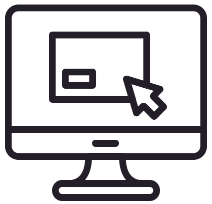
- Here, we have listed three components in the declarations array for AppModule.
- All these three components belong to AppModule.

Split an Application into Modules



User Module

Products Module



Root Module

Modules in Angular Application

UserModule

- ProfileComponent
- RegComponent

ProductsModule

- Products Component
- Cart Component

RootModule

- LoginComponent
- MenuComponent

Root Module and Feature Modules

Root and Feature Modules

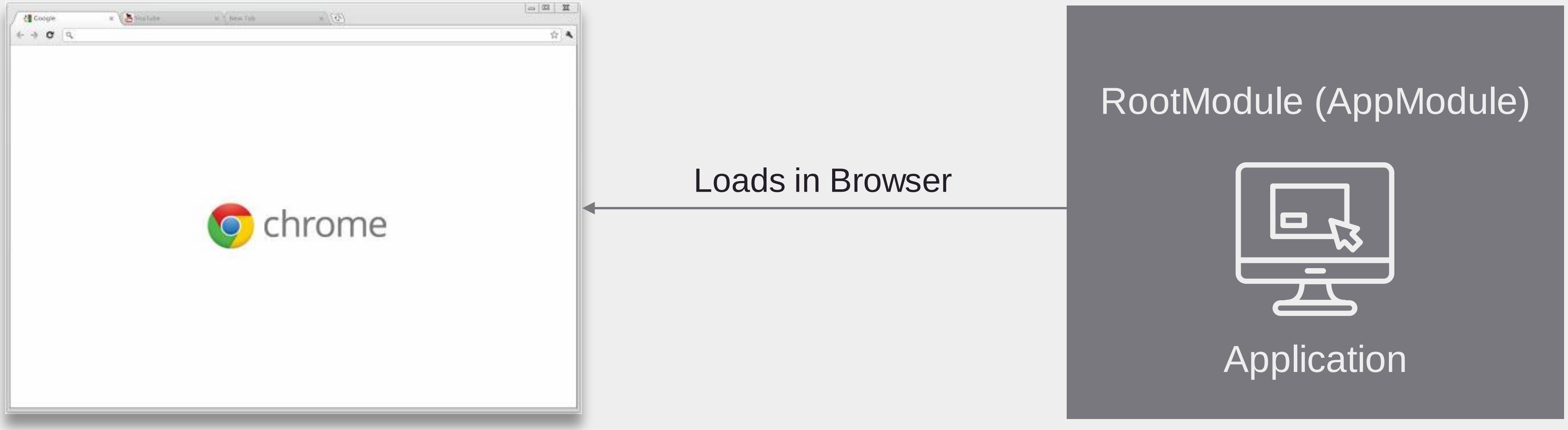
Root Module

- Used to load (bootstrap) an App.
- Only ONE Root module for every App.
- Generally named as AppModule.

Feature Modules

- An App can have feature modules based on functionality.
- Eg.: ProductsModule, AuthenticationModule

Loading Root Module in Browser



Bootstrapping of Root Module

- Bootstrapping refers to loading and initializing an application.
- Root Module is loaded/initialized in the main.ts file.

main.ts

src > main.ts > ...

```typescript

import { AppModule } from './app/app.module';
import { environment } from './environments/environment';

if (environment.production) {
  enableProdMode();
}

platformBrowserDynamic().bootstrapModule(AppModule)
  .catch(err => console.error(err));

```

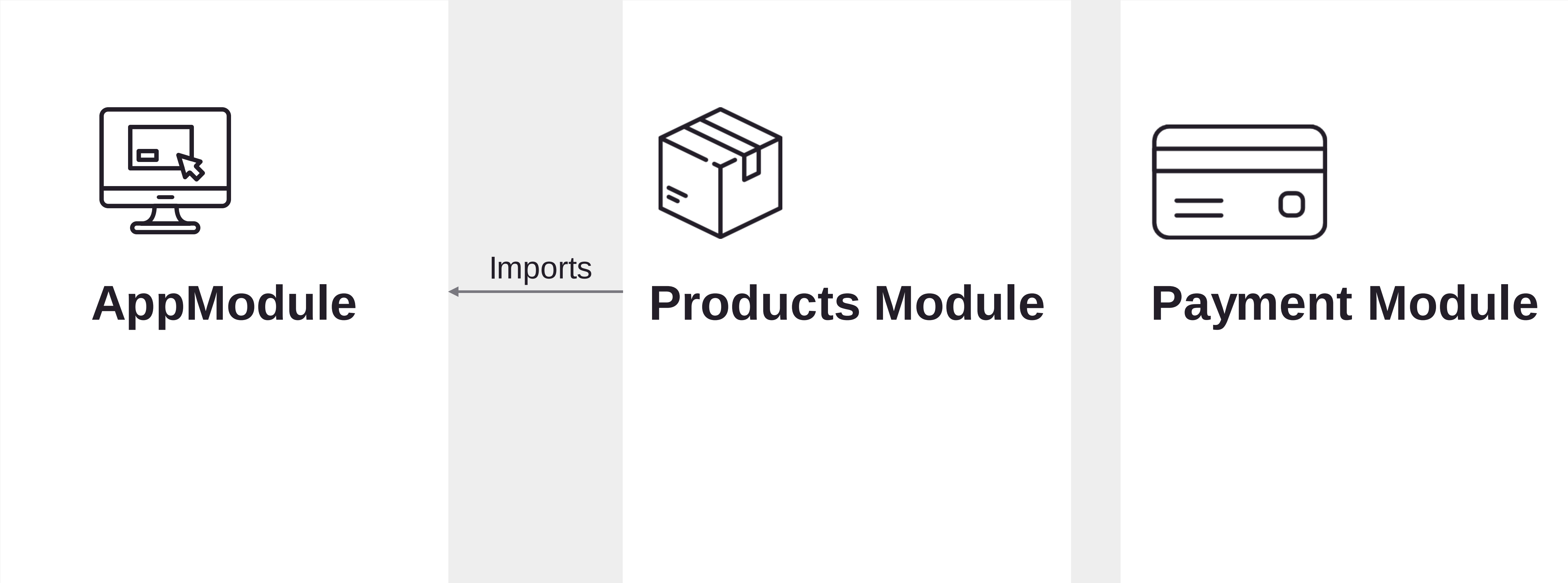
- The Angular application is bootstrapped with the AppModule.

Feature Modules

- An Application can be divided into various feature areas.
  - Customers
  - Authentication
  - Products
  - Users
- Each feature functionality can be developed as a Feature Modules.
- The Root Module will import the Feature Modules.

Root and Feature Modules

- An Angular Application has one **Root Module**.
- It can further have multiple **Feature Modules**.



Imports Array

- Angular is shipped as a set of modules.
  - BrowserModule (@angular/router)
  - HttpModule (@angular/common/http)
  - FormsModule (@angular/forms)
- Application can be made up of multiple modules as well.
  - App/UserModule
  - App/ProductsModule
- List all the required modules in the ‘imports’ array.

Imports Array vs Import Statements

- The JavaScript import statements give you access to classes/functions exported by other files so you can reference them within this file.
- The module's imports array tells Angular about specific Angular modules — Classes decorated with @NgModule — that the application needs to function properly.

```
imports: [
    BrowserModule,AppRoutingModule,FormsModule,HttpClientModule,
    CustomersModule,ReactiveFormsModule
  ],
```

TypeScript Modules != Angular2 Modules

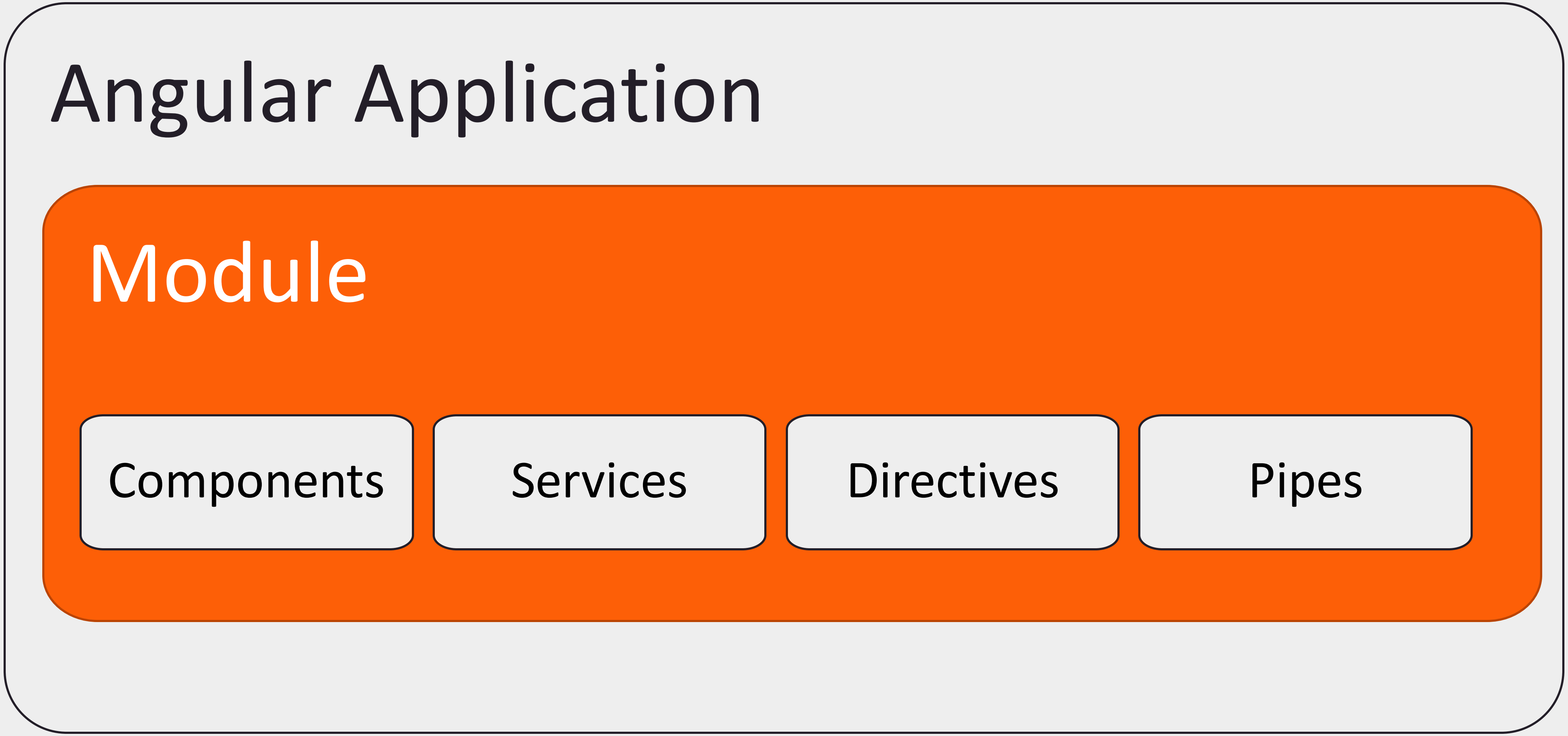
Angular Modules

An Angular Module groups together a set of Angular artifacts, namely components, directives, pipes and services that are part of that very same module.

TypeScript Modules

Whereas TypeScript Modules are a more generic way of programming in TypeScript.

Building Blocks of Angular Application



Summary

- What are Angular Modules
- Properties of @NgModule
- What is a Root Module
- Feature Modules

Appendix

References

Thank You

References

- https://angular.io/guide/ngmodules
- https://angular.io/guide/feature-modules
- https://www.tektutorialshub.com/angular/angular-modules/#how-to-create-an-angular-module
- https://www.tektutorialshub.com/angular/angular-modules/#ngmodule

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Thank you!

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